

Frank Lee

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EDUCATION

Carnegie Mellon University <i>M.S. in Electrical and Computer Engineering with Concentration in Embedded Systems</i>	Pittsburgh, Pennsylvania	Graduation date: May 2020 GPA 3.6/4.0
University of California, Davis <i>B.S. in Electrical Engineering with Concentration in Analog and Digital Circuits, Minor in Music</i>	Davis, California	Graduation date: June 2019 GPA 3.4/4.0

WORK EXPERIENCE

Cerebras Systems <i>Embedded Software Engineer</i>	Sunnyvale, California	Jan 2021 – Present
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Linux and Low-Level Programming:

- Engineering Linux daemons written in C and C++ that ranges from system-wide monitoring to Cerebras' unique wafer specific power and health monitoring, leveraging multithreading and shared libraries
- Developing all layers of the code, beginning from low-level communication (I2C, SPI) with the hardware, to library functions utilizing this communication for higher-level user level APIs
- Implementing communication protocols for IO-Link sensors, ensuring compliance with industry standards

Debugging and Signal Analysis:

- Utilizing oscilloscopes and logic analyzers to decode SPI/I2C communication signals, including UART-like protocols embedded within SPI signals (IO-Link)
- Using test fixtures and utility software to rigorously validate the behavior and performance of vendor-supplied components, such as sending raw I2C signals compliant with the PMBus protocol to test power modules

User Interfaces and APIs:

- Designing CLI and APIs in Golang, incorporating gRPC for seamless communication between different software components, thereby streamlining system administration tasks

Collaboration and Mentorship:

- Working collaboratively with design, manufacturing, and production teams to deploy dozens of systems monthly, leveraging expertise in software to troubleshoot hardware issues effectively during system bring-up
- Provided mentorship to an intern, facilitating their professional growth, acknowledging their achievements, and assisting them in securing a return offer

Yinzcam <i>Embedded Systems Engineer Intern / Hardware and Firmware lead (Athletech group)</i>	Pittsburgh, Pennsylvania	May 2020 – July 2020
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- Designed and created a proof-of-concept board capable of real-time biophysical trait measurement for athlete performance monitoring
- Developed a custom PCB using Autodesk Eagle software, incorporating a microcontroller, battery circuits, and 5-6 sensors, while meeting high-level functional requirements
- Configured pins on an ARM-based microcontroller to match required peripheral specifications, and wrote industry-standard embedded C firmware code using a modern IDE

Texas Instruments <i>Digital Design Engineer Intern (High Speed Signal Conditioning group)</i>	Santa Clara, California	June 2019 – August 2019
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- Designed and verified Verilog RTL code for integration with TI's latest PCIe Retimer chip

SKILLS

- Programming: C/C++, Embedded C, Python, Linux, GoLang (basic proficiency)
- Communication Protocols: I2C, SPI, UART, IO-link
- Software Applications: Git, Jira, VS Code
- Characteristics: Fast Learner, Punctual, Determined, Curious, Organized, Helpful, Patient